Car DJ - Reviewed

Yu Ting Huang Núria Molné Franquesa Adolfo Hernández Signoret

Abstract

We want to do a car-dj agent which starts up on 2 premises: your musical taste saved as a context, and an instruction which demands a specific type of feeling, this could be: "I want something happy" or "Play roadtrip vibes". Based on this voice instruction, it adds songs to the queue trying to make the driver match their emotion while also increasing driving safety. The driver receives 3 stimulus, the music being played, the interface of the Farhut bot which will display both music waves and gestures, and the questions asked by the Farhut bot to confirm his selections. The user will be able to reset his music choices by triggering the bot through a voice command, and the bot will try to identify how the user wants to feel, and then ask for confirmation on the emotion selected. During the whole session, the bot will also make the decision for the next song based on the emotions of the user perceived through images taken every x amount of time.

I. Introduction

Our system automatically suggests and plays songs based on how the driver feels. This way we avoid manual music selection while driving and, therefore, we remove the distraction this could cause. The system first asks the driver a few questions about how they are feeling, gets the answers through voice recognition, transfers them to text to analyze it to finally get the mood.

A. Schematic summary

1) Objectives
   1. Recognize the driver's emotion in real time, which we achieved training our own ResNet model for facial emotion recognition.
   2. Select music based on voice commands and detected emotions. We implemented it with a custom queue system that scores songs by emotion match.
   3. Create a natural and engaging interface, using Furhat robot gestures and visual feedback synchronized with music.

2) For training the image emotion recognition we used the FER-2013 dataset.

3) We tested with the FER pretrained model, then switched to ResNet18 for training our own.

4) Performance
   The emotion recognition had low latency, which for our goal is suitable for real-time music adaptation. Accuracy could be improved (future work). We also noticed some instability when we had variable lighting or head moving.
   For the voice recognition, the system detected correctly the emotion based on the driver's answers to the questions and it suggested songs accordingly.

5) We used the set of emotions: happy, neutral (or calm), angry, and sad.

6) Mapping schemes
   Each emotion influences the next song choice and Furhat robot gestures:
   - When emotion is set as happy, it suggests upbeat songs.
   - When the emotion is neutral, it suggests mildly positive songs, not as upbeat as happy.
   - When the emotion is angry, it suggests more harsh songs, like rock style. Future work: rather suggest calming songs to make the driver less angry.
   - When sad, it suggests uplifting songs.

7) Use of generative AI *(see below)*

8) Code Repository https://github.com/Sign0ret/cardj-reviewed
   - **Model Training Files:**
     - *Unused Models Comparison:* https://github.com/Sign0ret/cardj-reviewed/blob/master/MobileNetV2Densenet121.ipynb
     - *Final Model Training:* https://github.com/Sign0ret/cardj-reviewed/blob/master/ResNet.ipynb
     - *Demo Video:* https://github.com/Sign0ret/cardj-reviewed/blob/master/demo.mp4

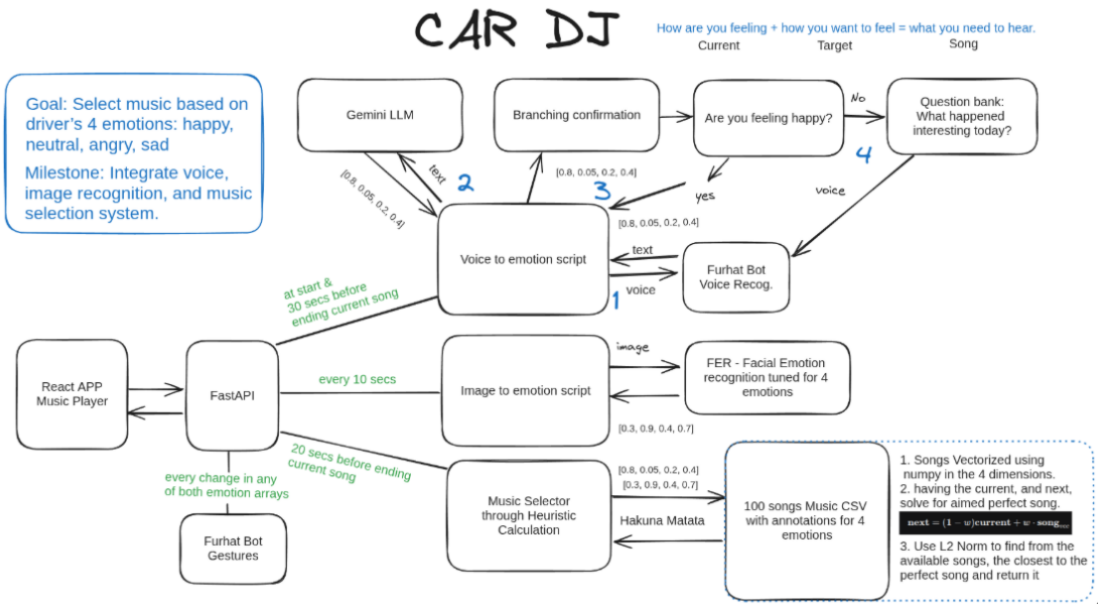

Fig. 1. Flow chart

B. Contribution statement

Yu Ting was responsible for voice recognition. Núria was responsible for documentation. Adolfo was responsible for image emotion recognition.

II. Methodology

A. Overall System Design

To achieve our goals, first, we decided to use the driver's saved music taste as a starting context, so the system already has a personal baseline instead of suggesting random songs. We also chose voice input as the main way for the driver to communicate with the system, as it is hands-free and helps reduce distraction while driving.

For emotion recognition through image, we chose the FER-2013 dataset, from which we focused on the set of emotions (happy, neutral, angry, and sad), as they are the ones most related to the moods a person can have. This also keeps the system fast and reliable, which is important for real-time use in a car.

We then trained a model (ResNet architecture) to recognize the emotion through facial image. Every 10 seconds, we analyze the expressions of the driver, to keep track of his emotions and see if their mood changes throughout the session. To manage music selection, we decided to build our own queue system. Songs are stored in a CSV file with emotion labels. This approach gives us control over how songs are scored and selected based on both the detected emotion and the voice commands.

For voice interaction, we decided to use the Furhat robot as the main interface. It handles voice recognition and engages the driver with short questions. We converted the voice input (driver's answers to the questions) to text and we sent it to the Gemini LLM to interpret the emotion. We also added a branching confirmation step (for example, asking "Are you feeling happy?") to reduce errors and improve confidence in the detected emotion.

The overall flow chart is shown in fig 1

B. Facial motion recognition system

1) Dataset: The dataset utilized for training is the FER-2013 dataset, comprising 35,887 grayscale images at 48×48 resolution. The data is partitioned into 80% training, 10% validation, and 10% testing sets. The class distribution is detailed in Table II. Since the emotion we need for car DJ application, we mapping to focused on the set of emotions (happy, neutral, angry, and sad) as the output of this submodule.

2) Model Selection and Comparison: To identify the most effective architecture, we benchmarked different CNN architectures representing distinct design philosophies. Initially, MobileNetV2 was compared against DenseNet121, which resulted to be a closer comparison to our target model, ResNet18.

MobileNetV2 (Efficiency Baseline): Designed for mobile optimization using depthwise separable convolutions. It serves as a benchmark for high-speed inference and training efficiency.

DenseNet121 (High-Performance Baseline): A model utilizing dense blocks to maximize feature reuse. While it offers performance comparable to ResNet, it typically requires more memory and computation time.

ResNet18 (Target Model): Our primary candidate, utilizing residual ”skip connections” to balance the deep feature extraction of DenseNet with the computational efficiency of MobileNet.

In this task, we focus on real-time implementation, where computation time should be as less as possible. All three models has similar accuracy. Therefore, we chose Resnet18 as our emotional detection model. The quantitative performance and efficiency metrics for these benchmark models are summarized in Table I in the appendix, as well as at figure 3, where the Confusion matrixes between MobileNetV2 and DenseNet121 models are shown side to side.

3) Training and Evaluation: The training results indicate a clear hierarchy in performance and efficiency. While MobileNetV2 consistently reached nearly 60% accuracy with much higher efficiency. DenseNet121 achieved a similar accuracy to MobileNetV2 but required significantly more computation time (213.16s).

This data provides a strong justification for the final selection of ResNet18, as it promises the robust feature extraction seen in DenseNet but with a more stable and efficient training process.The learning curve is shown in fig.4 in appendix.

C. Voice emotion detection system

1) Design: The main goal of the system is to improve human–robot interaction by enabling the robot to understand, confirm, and adapt to the driver’s emotional condition during driving. The flow of voice emotion detection system is shown in **??** in appendix

2) Implementation:

- Initialize Gemini prompt: The quality of the prompt has a significant impact on Gemini’s emotion recognition performance. Well-defined and specific prompts lead to more accurate and stable emotion probability outputs.

  Prompt Used

  You’re job is detecting the emotion of driver based on your conversation. In the end, reply a possibility of emotion(happy, calm, angry, sad). For example, when the driver says I had a bad day then the possibility might by [happy, calm, angry, sad] = [0.1, 0.3, 0.2, 0.4]. Return in json format with name emotion_possibility.

- Text analysis: The text analysis module combines both rule-based and LLM-based approaches. The rule-based method efficiently handles common and explicit responses, significantly reducing token usage, while the LLM-based approach is only applied when deeper semantic understanding is required. For example, in the emotion confirmation function, when the driver responds with uncertain expressions, the system must rely on longer and more complex conversational context to accurately analyze the driver’s emotional state.
- Robot Interaction: When Furhat speaks, it expresses corresponding facial expressions and gestures based on the conversational content, providing emotional support and enhancing the user’s interaction experience.

3) Results: The subsystem effectively detects and confirms driver emotions, especially for clear responses. Rule-based checks reduce LLM usage, but uncertain answers require longer dialogues and more processing. Overall, it performs well, though clarity of driver input affects efficiency.

D. Interaction Sub-system

1) Design: It assembles the python scripts for voice to text to emotional array and image to emotional array with the song recommendation system, all merged into a user interface to allow the user identify the top recommendations.

2) Implementation: To achieve the intended design, the dialogue management system does the following: through fastApi, all of the different scripts become endpoints consumable at different points during the interaction. First it triggers automatically the voice to text to emotional array script, when this is done, it now recommends songs with this initial data. Concurrently, the image to emotional array takes images every 10 seconds and adds the result to an average. With this 2 values in which the second is constantly rewritten, the songs get fetched instantly to match the current state of both the current emotion and target emotion. Through the React App, this 3 parts are displayed to the user to help him identify why he has been recommended this songs.

3) Results: As a result, the subsystem recommended successfully different types of songs according to which emotional target was detected. This then got also influenced by the current state of the user, and recommendations change as shown in the tests results in table III at the appendix.

III. General Discussion

A. Overall Pipeline

The pipeline performed well combining voice input, facial emotion recognition, and music selection. However, we would like the emotion recognition through image to be more important in the first song choice. It helps keep track of the driver's emotion without having to keep asking questions; however, it's not as important to set the emotion in a first instance.

B. Challenges Faced

The biggest challenge was understanding how to train the models, and how to identify which one had a better potential of being trained with the chosen data. Also getting to work with the Furhat API and making it work with the movements of the head, aiming for the greatest interactivity with the user in real time was hard, so we chose to use different movements, in a branching way, for each detected emotion in the voice prompts given by the user.

C. Use of Generative AI

In order to get the best training solutions, the usage of the google colab environment and the use of Google Gemini as helper for bugs and debugging where used in this environment.

D. Ethical Issues

One of the main ethical concerns in this project is the privacy. The system uses personal data (driver's facial expressions and voice). Although this data is only used to improve the driving experience, it could be misused if it was stored, shared, or accessed without the user's consent. To address this, we would have to inform users about what data is being collected and processing emotion data only locally (ideally), without saving raw images or audio.

IV. Conclusion and Future Work

To develop a working solution that uses 2 sources of input including computer vision, as well as voice-to-text, and try to obtain certain output based on the emotions recognized in both of them, required to pass through several processes, including the training and benchmarking of different models such as ResNet18, MobileNetV2 and DenseNet121, as well as the implementation of the furhat bot api and the usage of a middle-man to facilitate some parts of the work, which was gemini api for us, a demo version for an interactive intelligent system that works with the 4 emotions: happy, sad, calm and angry was achieved and was used to interact with the user via songs recommendations. Also, after executing the demo, not having trained the model on our own labeled images made it harder for it to detect the emotions of Adolfo, the subject of the demo, but this also allowed to know how overfiting, even though it might display great results in demos, might not be the best solution for general-use products. In order to make it better, training it on a larger dataset with smaller gestures rather than exaggerated ones should make it more accurate.

For future work, improvements in the furhat interactive responses could be added, as well as a faster input digestion in the real-time demo.

References

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APPENDIX

Voice emotion flow chart

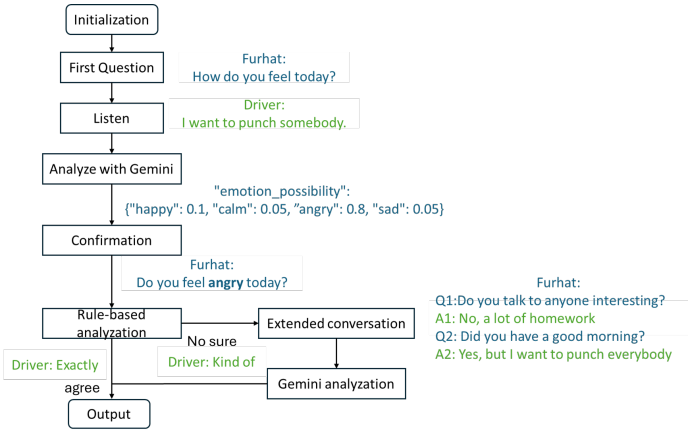


Fig. 2. Voice emotion detection flow chart

Emotion recognition through images. Model training results.

Model	Train Size	Comp. Time (s)	Test Acc	W-F1
Initial Benchmark:				
Resnet18	28,709	100.11s	56.23%	0.571
MobileNetV2	28,709	128.43s	59.91%	0.60
DenseNet121	28,709	213.16s	59.93%	0.60

TABLE I

EFFICIENCY AND PERFORMANCE METRICS COMPARING BENCHMARK MODELS ACROSS TRAINING ITERATIONS.

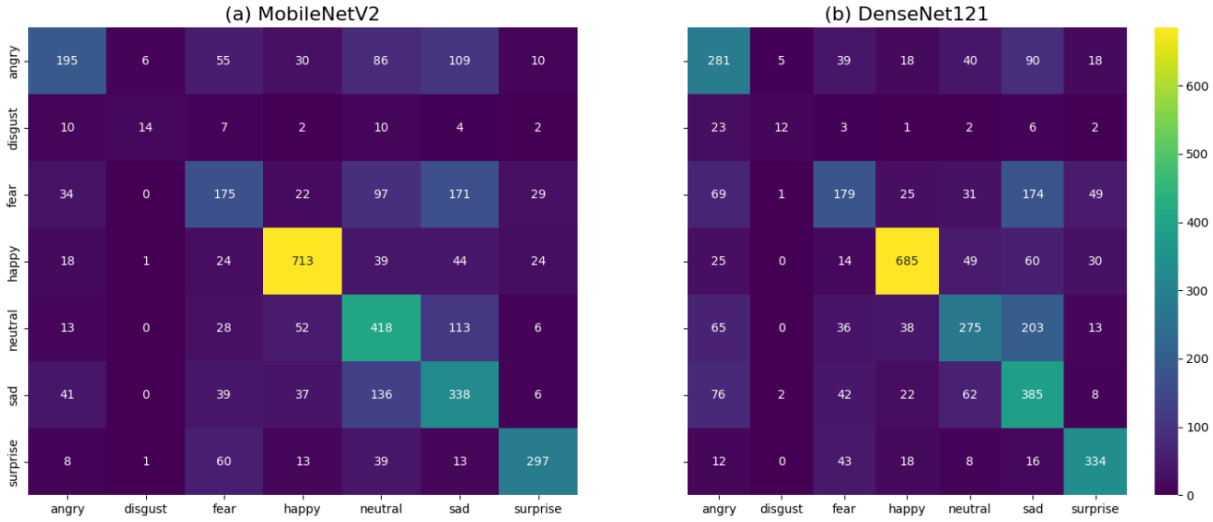


Fig. 3. Confusion matrix comparison generated during benchmarking: (a) MobileNetV2 and (b) DenseNet121.

Emotion	Train	Public Test (Val)	Private Test
Angry	3995	467	491
Disgust	436	56	55
Fear	4097	496	528
Happy	7215	895	879
Sad	4830	653	594
Surprise	3171	415	416
Neutral	4965	607	626
Total	28,709	3,589	3,589

TABLE II
OFFICIAL DATA SPLIT AND CLASS DISTRIBUTION OF THE FER-2013 DATASET.

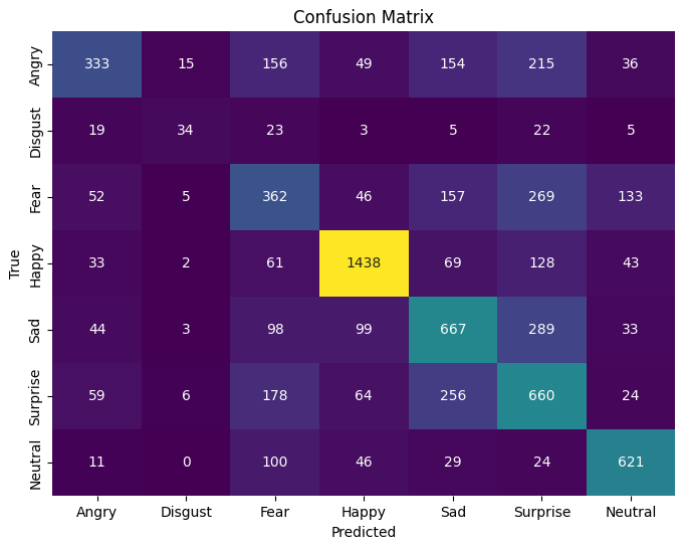


Fig. 4. Resnet confusion matirx

Song recommendation results for different emotions.

Target	Current	Song Recommended #1	Hap	Sad	Ang	Cal
Sad	Happy	Stick Season - Noah Kahan	0.3	0.6	0.1	0.5
Happy	Happy	Million Dollar Baby - Tommy Richman	0.8	0.1	0.1	0.1
Calm	Happy	I Remember Everything - Zach Bryan	0.3	0.7	0.1	0.4
Angry	Happy	Not Like Us - Kendrick Lamar	0.2	0.1	0.9	0.0

TABLE III
EMOTION-BASED SONG RECOMMENDATIONS WITH INTENSITY SCORES

Project's specifications (next page):

Car-DJ

Yu Ting Huang Adolfo Hernández Signoret Núria Molné Franquesa

I. Introduction

We want to do a car-dj agent which starts up on 2 premises: your musical taste saved as a context, and an instruction which demands a specific type of feeling, this could be: ”I want something happy” or ”Play roadtrip vibes”. Based on this voice instruction, it adds songs to the queue trying to make the driver match their emotion while also increasing driving safety. The driver receives 3 stimulus, the music being played, the interface of the Farhut bot which will display both music waves and gestures, and the questions asked by the Farhut bot to confirm his selections. The user will be able to reset his music choices by triggering the bot through a voice command, and the bot will try to identify how the user wants to feel, and then ask for confirmation on the emotion selected. During the whole session, the bot will also make the decision for the next song based on the emotions of the user perceived through images taken every some amount of time.

II. Elevator Pitch

We want to build a Car agent to customize the music playlist and keep the driver safe and satisfied with music.

III. Objectives

1) **Objective 1:** Music Taste analysis: Build a music list to simulate the driver´s personal music library and label their corresponding emotion manually. We will use Excel to store the list of music and process the data with pandas.
2) **Objective 2:** Emotional recognition: Each x seconds before the playing song ends, trigger the emotional recognition model with phased frames by y seconds to cover the different emotions through these final x seconds, leaving enough margin for the music selection system. The model we plan to use is OpenCV for landmark extraction, since we are aiming forreal-time implementation, the low latency is the first priority for choosing. ResNet18 PyTorch model fine tuning it for the following emotions: ”angry”, ”happy”, ”sad”, ”calm”. In order to achieve this, the DiffusionFER dataset provided for the first lab will be used.
3) **Objective 3:** Music selection based on inputs: Having the average of emotions recognized and the last received conversation through the voice commands between the farhut bot and the user, it finds the suitable song based on the sum of the inputs received and adds it to the queue. In order to achieve this, and since the focus of the project is emphasized on the Farhut interaction, the music will be manually classified for each of the four emotions used previously.
4) **Objective 4:** Robot interface and the corresponding reaction with music (dance or wave). To accomplish this goal we list all the robot gestures with FurhatRemoteAPI.
5) **Objective 5:** At the final seconds of the playing song, the robot asks for the user’s desired emotions again and suggests another music based on the conversation or the emotion detected during the previous song.

IV. Deliverables

1) **Deliverable 1:** Dataset for music with emotion tags. Deliver a musicList.csv and explain the standard of how we label them (based on emotion and safety).
2) **Deliverable 2:** Datasets for emotions (training and testing) in csv format and training for emotion recognition model.
3) **Deliverable 3:** Music player, with current song and next song picking functionality.
4) **Deliverable 4:** Finite state machine for knowing when to trigger the image recognition for emotions, when to get the next song and when to play the next song.
5) **Deliverable 5:** Video Demo: Real-time implementation and make sure the detection is correct and the music transition is seamless between songs.

V. Success Metrics

- **Metric 1:** Music dataset completeness: should have the same amount of samples of each 4 emotions (happy, sad, angry, calm).
- **Metric 2:** Testing accuracy should be larger than 70%.
- **Metric 3:** The robot can successfully acquire relevant emotional targets through voice conversations with the user.
- **Metric 3:** The robot can do 4 different movements.
- **Metric 4:** Video demo should include all the states of the FSM.

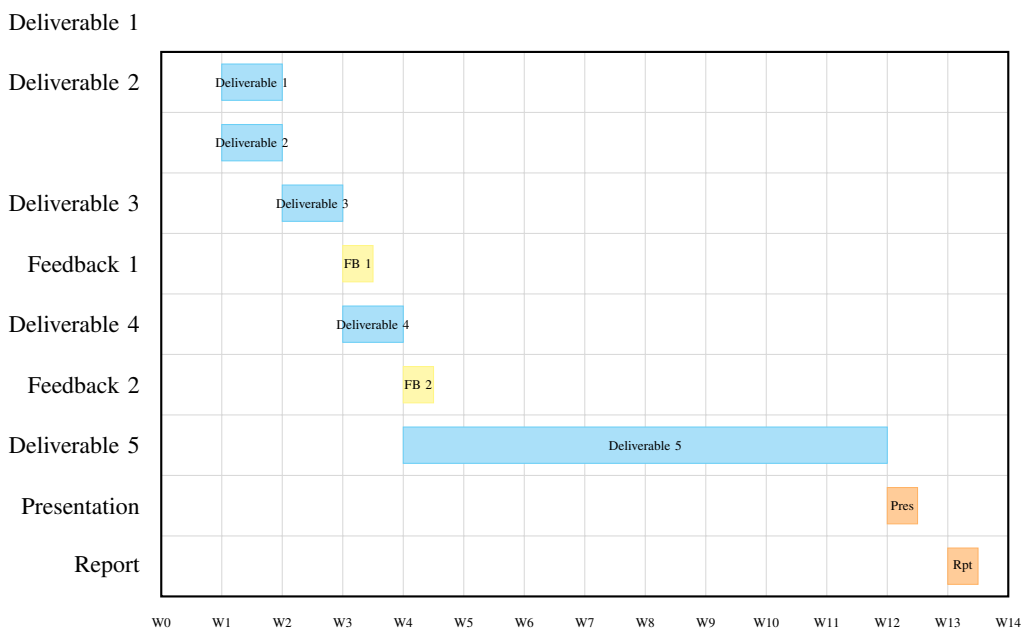
VI. Potential Issues

- **Issue 1:** Underfitting or overfitting of the model in training.
- **Issue 2:** FSM correct implementation.
- **Issue 3:** Correct measurement of emotions in real life due to only taking samples of frames during the song rather than the whole period.
- **Issue 4:** The computation duration (landmark detection and ResNet18) is too long.

VII. Project Breakdown

Deadline	Task Description	Assigned To	Notes/Dependencies
November 30	Deliverable 1	Adolfo	Classified by 4 emotions, each having a value of 0-1
November 30	Deliverable 2	Núria	Reliable dataset into 4 emotions and train the model
December 07	Deliverable 3	Adolfo and Yu Ting	Voice Recognition, LLM usage
December 07	Deliverable 4	Adolfo	3 systems (voice, image, music) synchronized
December 09	Feedback Session 1	Team members (all)	Deliverables 1-4
December 14	Deliverable 5	Adolfo and Yu Ting	Display 3 working systems
December 18	Feedback Session 2	Team members (all)	Deliverable 5
January 14	Project Presentation	Team members (all)	Real-time demonstration
January 16	Report Submission	Team members (all)	

Gantt Chart:



Legend: Task Feedback/Checkpoint Milestone

VIII. Aimed Grade

Grade 5, The idea of a car DJ has now been transformed into objectives, these objectives satisfy the requirements given by applying the abilities learned to train a model, orchestrate a working demo environment to display the use of the solution, and use cutting edge self hosted models for voice recognition, as well as the API of an LLM for song picking for more natural to interact with social robots and versatile in use.